GORENSTEINNESS OF THE LOCAL RING OF A FIBRE IS ETALE LOCAL ON SOURCE AND TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6483-6494 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6496-6512, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Gorenstein local rings, local rings of a fibre, and etale locality of properties of morphisms at a point; the comparison is retained.

Source reference. Original lemma 00U4: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6512–6512. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6512–6512. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NI: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6512–6512. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BJL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6512–6512. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AGZ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6512–6512. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *The property of morphisms of germs of schemes*

$$\mathcal{P}((X, x) \to (S, s)) =$$
$$\text{the local ring } \mathcal{O}_{X_s,x} \text{ of the fibre is Noetherian and Gorenstein}$$

is étale local on the source-and-target (Descent, Definition 04NB).

Proof. Given a diagram as in Descent, Definition 04NB we obtain an étale morphism of fibres $U'_{v'} \to U_v$ mapping u' to u, see Descent, Lemma 04NI. Thus $\mathcal{O}_{U_v,u} \to \mathcal{O}_{U'_{v'},u'}$ is the localization of an étale ring map. Hence the first is Noetherian if and only if the second is Noetherian, see More on Algebra, Lemma 0AGZ. Then, since $\mathcal{O}_{U'_{v'},u'}/\mathfrak{m}_u\mathcal{O}_{U'_{v'},u'} = \kappa(u')$ (Algebra, Lemma 00U4) is a Gorenstein ring, we see that $\mathcal{O}_{U_v,u}$ is Gorenstein if and only if $\mathcal{O}_{U'_{v'},u'}$ is Gorenstein by Dualizing Complexes, Lemma 0BJL. □